(3 0 2 0 2 0 0 1 1 4) = 13/26

CLASS RANK: 7/18

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Assignment #1:

AN INTRODUCTION TO THE WORLD OF MATHEMATICAL PROOF

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ABSTRACT. In this article, we explain the history of algebraic and combinatorial mathematical proofs, what they are, and the application and use of them in the modern day world, and in the world of mathematics.

1. INTRODUCTION

A *proof* is an **explanation** of why a statement is **objectively correct**, simply meaning, a proof is why a statement is correct, no matter the circumstances. There are two ways of going about this, proving its *veracity* and provide the *exposition*. Veracity is verify if the statement is objectively correct, and exposition is the explanation to why the proof is correct. There are many ways of doing this, but there are two main types of proofs, algebraic and combinatorial.

Algebraic proofs utilizes a Formal Proof System. A Formal Proof System consists of 1. A set of expressions called *statements* and 2. A set of rules called *rules of inference*. Also defined are axioms, which are conclusions of a rule of inference that has no premises. Next, A statement Q in a formal axiom system is *provable from* premises $Q_1, Q_2, ..., Q_n$ if: 1. Q is a conclusion of a rule of inference when $P_1, ...P_k$, are the premises, and 2. for each $1 \leq i \leq k$, if P_i is provable from $Q_1, ..., Q_n$. Fourth, if Q follows from no premises in a formal axiom system, we say that Q is *provable* in the system. A provable statement is called a *theorem*. Lastly, a *proof* of a statement in a formal axiom system is a sequence of applications of the rules of inference (i.e., *inferences*) that show that the statement is a theorem in that system. All of these together allow for the wide range of algebraic proofs. Three forms of algebraic proofs include Formal, Semiformal and Expository. Formal proofs list every reason and every sequence of reasons together, but there are no shortcuts allowed. When a proof is proved, it can be referenced at any time, because it is objective truth, but in a formal proof, when using a shortcut, you must list every possible step through it. Next down the line is semiformal proofs, which is the midpoint between formal and expository. semiformal proofs allow the use of previously formed theorems. If a theorem has been proved as objectively true, then it can be applied to anything, even other theorems. Lastly, expository proofs are when English meets math.

When writing expository proofs, proper English grammar, and to provide a good exposition to the reader.

On the other hand are Combinatorial proofs. A *combinatorial proof* (or *combinatorial argument*) is proof of a combinatorial identity obtained by counting the same thing (or two equivalent things) in two different ways. These proofs are usually more intuitive, visually and conceptually easier to wrap your brain around. These proofs can deal with buckets of water, or people in groups, or any list of things that describes counting something twofold. These proofs are how little kids learn addition. A teacher shows two cookies, and brings two more cookies together, and the students all shout out "FOUR COOKIES!!". The proof for 2 + 2 is ridiculous to teach to a five-year-old, but showing them a combinatorial proof is much simpler and intuitive.

In this paper, the three types of Algebraic proofs (formal, semiformal, and expository) and Combinatorial proofs.

[1]

2. Pierce's Law | Formal Proofs

This axiom, Pierce's Law, was discovered by a philosophy by the name Charles Sanders Pierce. This can also be used similarly to the law of excluded middles. but written in a form that involves only one sort of connective, namely implication. [2]

We can ONLY prove tautologies in Prop Logic

In Propositional Calculus, *Pierce's Law* says that $((P \Rightarrow Q) \Rightarrow P) \Rightarrow P$, saying that, If there is a boolean Q such that the true value of P stems from the true value of "if P then Q" that P must be true. Because of this, Pierce's Law implies the law of Excluded Middle

This proof, however seems like it is a tautology, because either $(P \Rightarrow Q) \Rightarrow P$ is either false or true, when this statement implies P, then it is still true. Although it seems so, $[Z \Rightarrow X] \Rightarrow [((P \Rightarrow Q) \Rightarrow Z) \Rightarrow P$ *IS a tautology*.

Below is the proof for Pierce's Law, using various other laws, such as the Proof by cases, DeMorgan's Law (2) ($\neg(P$ or $Q) \Rightarrow \neg P$ and $\neg Q$), Proof by contrapositive ($(P \Rightarrow Q) \Leftrightarrow (\neg Q \Rightarrow \neg P)$) and lastly, Excluded Middle, ($\neg P$ or P).

Theorem (4.19 Pierce's Law)**.** $((P \Rightarrow Q) \Rightarrow P) \Rightarrow P$

Proof.

1.	Assume $(P \Rightarrow Q) \Rightarrow P$	-
2.	Assume $\neg((P \Rightarrow Q)$ or $\neg(P \Rightarrow Q))$	-
3.	Assume $P \Rightarrow Q$	-
4.	$(P \Rightarrow Q)$ or $\neg(P \Rightarrow Q)$	by or+; 3
5.	$\rightarrow\leftarrow$	by $\rightarrow\leftarrow$ +; 2, 4
6.	$\neg(P \Rightarrow Q)$	by not-; 3, 5
7.	$(P \Rightarrow Q)$ or $\neg(P \Rightarrow Q)$	by or+; 6
8.	$\rightarrow\leftarrow$	by $\rightarrow\leftarrow$ +; 2, 7
9.	$(P \Rightarrow Q)$ or $\neg(P \Rightarrow Q)$	by not-; 2, 8
10.	$\leftarrow$	-
11.	$(P \Rightarrow Q) \Rightarrow P$	by copy; 1
12.	Assume $\neg(P \Rightarrow Q)$	-

13.	Assume $\neg P$ or Q	-
14.	Assume $\neg P$	-
15.	Assume $\neg Q$	-
16.	$\leftarrow$	-
17.	$\neg Q \Rightarrow \neg P$	by $\Rightarrow$ +; 13, 14
18.	$\leftarrow$	-
19.	Assume $\neg Q \Rightarrow \neg P$	-
20.	Assume P	-
21.	Assume $\neg Q$	-
22.	$\rightarrow\leftarrow$	by $\rightarrow\leftarrow$ +; 13, 17
23.	$\leftarrow$	-
24.	Q	by not-; 18, 19
25.	$\leftarrow$	-
26.	$P \Rightarrow Q$	by $\Rightarrow$ +; 17, 20
27.	$\leftarrow$	-
28.	$(\neg Q \Rightarrow \neg P) \Rightarrow (P \Rightarrow Q)$	by $\Rightarrow$ +; 16, 21
29.	$P \Rightarrow Q$	by $\Rightarrow$ −; 15, 22
30.	$\neg P \Rightarrow (P \Rightarrow Q)$	by $\Rightarrow$ +; 13, 23
31.	Assume Q	-
32.	Assume P	-
33.	$\leftarrow$	-
34.	$P \Rightarrow Q$	by $\Rightarrow$ +; 25, 26
35.	$\leftarrow$	-
36.	$Q \Rightarrow (P \Rightarrow Q)$	by $\Rightarrow$ +; 25, 27
37.	$P \Rightarrow Q$	by or-; 12, 24, 28
38.	$\rightarrow\leftarrow$	by $\rightarrow\leftarrow$ +; 11, 29
39.	$\leftarrow$	-
40.	$\neg(\neg P$ or $Q)$	by not+; 12, 30
41.	Assume $\neg(\neg P$ or $Q)$	-
42.	Assume $\neg(\neg\neg P$ and $\neg Q)$	-
43.	Assume $\neg P$	-
44.	$\neg P$ or Q	by or+; 34
45.	$\rightarrow\leftarrow$	by $\rightarrow\leftarrow$ +; 32, 35
46.	$\leftarrow$	-
47.	$\neg\neg P$	by not+; 34, 36
48.	Assume Q	-
49.	$\neg P$ or Q	by or+; 38
50.	$\rightarrow\leftarrow$	by $\rightarrow\leftarrow$ +; 32, 38

+3

+1 cases

51.	$\leftarrow$	-
52.	$\neg Q$	by not+; 38, 40
53.	$\neg\neg P$ and $\neg Q$	by and+; 37, 41
54.	$\rightarrow\leftarrow$	by $\rightarrow\leftarrow$ +; 33, 42
55.	$\leftarrow$	-
56.	$\neg\neg P$ and $\neg Q$	by not-; 33, 43
57.	$\leftarrow$	-
58.	$\neg(\neg P$ or $Q) \Rightarrow \neg\neg P$ and $\neg Q$	by $\Rightarrow$ +; 32, 44
59.	$\neg\neg P$ and $\neg Q$	by $\Rightarrow$ $-$; 31, 45
60.	$\neg\neg P$	by and-;46
61.	Assume $\neg P$	-
62.	$\rightarrow\leftarrow$	by $\rightarrow\leftarrow$ +; 47, 48
63.	$\leftarrow$	-
64.	P	by not-; 48, 49
65.	$\leftarrow$	-
66.	$\neg P \Rightarrow (P \Rightarrow Q) \Rightarrow P$	by $\Rightarrow$ +; 11, 50
67.	P	by or-; 9, 10, 51
68.	$((P \Rightarrow Q) \Rightarrow P) \Rightarrow P$	by $\Rightarrow$ +; 1, 52

□

3. Induction | Semiformal Proof

Induction is a fundamental cornerstone for proving proofs. When proving for induction, there is a three-step method for proving said proofs. First, the base case. This case proves that starting at the base of the equation, the equation is true. Explaining induction with dominoes is the best way. The base case is seeing if the first domino falls and hits the second domino. Next comes the hypothesis step, where you substitute a variable for k, and assume that it is equal to the original, for example, Gauss's formula shown below states;

$$1 + 2 + 3 + ... + n = \frac{n(n+1)}{2}$$

translates to

$$1 + 2 + 3 + ... + k = \frac{k(k+1)}{2}$$

Following the Hypothesis step, is the Inductive step. To prove this, take the value of k and simplify the equation stated in the Hypothesis step, and prove that it increases for the amount that k increases

$$1 + 2 + 3 + ... + k + (k+1) = \frac{(k+1)((k+1)+1)}{2} = \frac{(k+1)}{2}(k+2)$$

then prove that adding $(k+1)$ to the normal formula:

$$= \frac{(k(k+1))}{2} + (k+1) = \frac{k(k+1)}{2} + \frac{2(k+1)}{2} = \frac{(k+1)}{2}(k+2)$$

Because these two equations are identical, we have proved that $1 + 2 + 3 + ... + n = \frac{n(n+1)}{2}$ is true, by proof by induction. By proving that $P(k+1)$ holds true, when proving that any domino, when pushed over, makes the one following it fall, showing that it is true for all of the dominoes.

Theorem (7.9 Zero Sums). $\forall m, n, m + n = 0 \Rightarrow m = n = 0$

Proof. Let m be arbitrary

Assume $m + 0 = 0$	
$m = 0$	by A0
$0 = 0$	by Definition of Reflexive
$m = 0$ and $0 = 0$	by Definition of And
$\leftarrow$	
$m + 0 = 0 \Rightarrow m = 0$ and $0 = 0$	by Definition of $\Rightarrow$
Let ~~n~~ k be arbitrary	
Assume $m + k = 0 \Rightarrow m = k$ and $k = 0$	
Assume $m + \sigma(k) = 0$	
$0 = m + \sigma(k)$	by Commutative of Equals
$= \sigma(m + k)$	by A1
Assume $0 \neq \sigma(k)$	
$\forall n,\ 0 \neq \sigma(n)$ — there is no 'n'	by N3
$m + k = m + k$	by Reflexive
$0 \neq \sigma(m + k)$	by Substitution
$0 = \sigma(m + k)$	by Copy
$\rightarrow\leftarrow$	by $\rightarrow\leftarrow$
$\leftarrow$	
$= \sigma(k)$	by Proof by Contradiction
$= 0$	by Reflexive
$= m$	by Substitution
$m = \sigma(k)$ and $\sigma(k) = 0$	by Definition of And
$\leftarrow$	
$m + \sigma(k) = 0 \Rightarrow m = \sigma(k)$ and $\sigma(k) = 0$	by Definition of $\Rightarrow$
$\leftarrow$	
$\forall n, m + n = 0 \Rightarrow m = n$ and $n = 0$	by Induction
$\leftarrow$	
$\forall m, \forall n, m + n = 0 \Rightarrow m = n = 0$	by $\forall+$

$\square$

don't insert a proof in the middle of a transitive chain

No ... how do you know this? you would need to $\Rightarrow$ - on the assumption

$+0$

4. Asymmetry | Semiformal Proof

Inequalities are an essential part to modern day computing, which is my field of expertise. Booleans, which utilize inequalities heavily, rely on the output of less than ($x < y$), greater than ($x > y$), equals ($x = y$). not equals ($x \neq y$ or x != y), and many more. These booleans help the flow of programming, for example in iterating through a loop, and for different output cases when needed. The proof below, if $x < y$ then $y \not< x$, is a great example of a boolean value, where if x is not less than y, y is either greater than, or equal to x, which is the basis of the proof.

Theorem (9.17 Asymmetry). *if $x < y$ then $y \not< x$*

Proof.

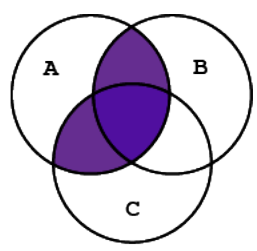
+2

Assume $x < y$	
Assume $y < x$	
~~$x < y$ and $y < z \Rightarrow x < z$~~	~~the Definition of Transitive~~
$x < y$ and $y < x \Rightarrow x < x$	by ~~Substitution~~ def of transitive
$x < x$	by Definition of $\Rightarrow$
$\neg(x < x)$	the ~~Definition of Reflexive~~ irreflexivity of $<$
$\rightarrow\leftarrow$	by $\rightarrow\leftarrow$
$\neg(y < x)$	by Definition of Contradiction
$y \not< x$	by Definition of $<$
$x < y \Rightarrow y \not< x$	by Definition of $\Rightarrow$

□

5. Distributivity of ∩ over ∪ | Semiformal Proof

The Law of Distributivity is a very universal law, that can be applied to algebra (addition, multiplication, etc.), probability, matrices, and proven below, sets. When looking at the relation between sets, visualizing how a set works instead of just letters helps a lot. Below is an image of the set $A \cup (B \cap C)$, where the area in A that overlaps with either B or C is shaded in. How the distributivity law of sets work in this equation is that A AND $B \cup C$ are colored in, where either B OR C can be used. on the contrary, if the equation were to be $B \cap C$, the darker purple in the middle would be shaded in, because what is contained in both $A \cap C$ AND $A \cap B$ is the shaded part in the middle. Below is the proof for $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.



A AND (**B** OR **C**)

+1 image

[4]

Theorem (10.23 Distributivity of ∩ over ∪). $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Proof.

No... you can't have a set equal to a statement

Let $A \cap (B \cup C)$ be a set	
$A \cap (B \cup C) = x \in A$ and $x \in B \cup C$	by Definition of Intersection
$= x \in A$ and $(x \in B$ or $x \in C)$	by Definition of Union
$= (x \in A$ and $x \in B)$ or $(x \in A$ and $x \in C)$	by Law of Distribution (4.38)[3]
$= (x \in A \cap B)$ or $(x \in A \cap C)$	by Definition of $\cap$

$= x \in (A \cap B) \cup (A \cap C)$ by Definition of $\cup$

□

6. Composition of Injective is Injective | Semiformal and Expository Proof

Composition of sets are essentially eliminating the middle man transition from one set to another. A composition says that, if $A \to B$ and $B \to C$, then $A \to C$. This law is similar to Transitivity, but has to do with the relation of the first set, compared to the relation in the second set. Injection is the relation of A to B, where every element in A, has a relation in B, but not every element in B, has a relation coming from A, meaning there are more B values than A values. In this problem, we prove that Injection is closed under the Law of Transitivity.

Theorem (01.47 Composition of injective is injective). *$f : A \to B$ and $g : B \to C$ are injective $\Rightarrow g \circ f$ is injective*

Proof.

Assume $f : A \to B$ and $g : B \to C$ are injective

$(g \circ f) : A \to C$ by Definition of Composition

Let $x, y \in A$

Assume $g(f(x)) = g(f(y))$

~~$g(x) = g(y) \Rightarrow f(x) = f(y)$ by Definition of Injective~~

x2 ce

$f(x) = f(y)$ by ~~Definition of $\Rightarrow$~~ injective

~~$f(x) = f(y) \Rightarrow x = y$ by Definition of Injective~~

$x = y$ by ~~Definition of $\Rightarrow$~~ injective

$(g \circ f)$ is injective by Definition of Injective

$f : A \to B$ and $g : B \to C$ are injective $\Rightarrow g \circ f$ is injective by Definition of $\Rightarrow$

□

Proof.

First, assume that $f : A \to B$ and $g : B \to C$ are injective. Next, by Definition of Composition, $(g \circ f) : A \to C$ and $(g \circ f)(x) = g(f(x))$. Let x, y be elements of A, and Assume that $g(f(x)) = g(f(y))$. Following, if $g : B \to C$, g is injective, and $g(x) = g(y)$, so $(g(f(x)) = g(f(y))) \Rightarrow (f(x) = f(y))$ by the definition of injective. Similarly, if $f : A \to B$, f is injective, and $f(x)$ and $f(y)$, then $f(x) = f(y) \Rightarrow (x = y)$ by the definition of injective. Ultimately, if x equals y, then $(g \circ f)$ is injective, by the definition of injective

□

7. Carl Friedrich Gauss | Combinatorial Proof

Carl Friedrich Gauss was a German Mathematician who made significant contributions to many fields in mathematics and science. Gauss was so mart, that at age 5, when Gauss was in elementary school, his teacher asked the class what 1 + 2 + 3 + ... + 100 came out to be. He had the idea of taking the sum,

$$\cancel{\sum_{i=1}^{n} \frac{n + (n+1)}{2}}$$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Which ends up being the sequence adding to 100 previously stated. Albeit this equation was found before Gauss was born, around the time of Archimedes and Pythagoras. He had the idea of taking the summation and add it to itself, but reverse the order. Coincidentally, the relation between the two sums turns into $101 + 101 + 101 + ... + 101$. Now, instead of manually adding all 100 number together, it can be simplified into $101 \cdot 100$, and because we added the sum to itself, $2 \cdot (101 \cdot 101) = 101 \cdot 50 = 5050$, the solution to $1 + 2 + 3 + ... + 100$. Gauss is also know for other formulas, for example, Physics. Another equation he created states that the total electric flux out of a closed surface is equal to the charge enclosed divided by the permittivity. The electric flux in area is defined as the electric field multiplied by the area of the surface projected in a plane and perpendicular to the field:

$$\oint \vec{E} . \vec{d}s = \frac{1}{\epsilon_0} q$$

:)

[6]

This formula is way over my head, and writing it in latex is fun, but it is an interesting equation nonetheless. One sequence that Gauss formed was the

$$3 \cdot (1^2 + 2^2 + 3^2 + ... + (n-1)^2 + n^2) = (2n+1) \cdot \binom{n+1}{2}$$

which is stated below as a combinatorial proof:

Theorem (12.29 Sum of Squares). $3 \cdot (1^2 + 2^2 + 3^2 + ... + (n-1)^2 + n^2) = (2n+1) \cdot \binom{n+1}{2}$

Proof. Imagine that you are in a kitchen, where there are three identical sets of n cookie jars. For each cookie jar starting at n, place n quantities of n cookies to the desired jar. Then, for every consecutive jar, place $(n-1)$ quantities of $(n-1)$ cookies from the previous jar, until there is one jar left. Lastly, count the total amount of cookies in all of the jars combined.

$$3 \cdot (n^2) + 3 \cdot ((n-1)^2) + (3 \cdot (n-1) - 1^2 = 3 \cdot (n-2)^2) + ... + 3 \cdot 2^2 + 3 \cdot 1^2$$

Simplifies to: simplification is NOT a combinatorial proof... it is algebraic

$$3 \cdot (1^2 + 2^2 + 3^2 + ... + (n-1)^2 + n^2)$$ ← just count this directly

Also in the kitchen are n jars, and Mom brings in n jars as well, and one special jar with cookies also inside of it. In each of these jars, you place n cookies, and one extra in each for good luck. Then for each jar, count how many combination of cookies in each jar $n+1$ when you can only choose two from the jar. Next, for each jar, count the amount of combinations you can make, and combine the total. This total is equivalent to the total amount of cookies in the previous combination.

$$(2 \cdot n + 1) \cdot \binom{n+1}{2}$$

+0

Meaning that:

$$3 \cdot (1^2 + 2^2 + 3^2 + ... + (n-1)^2 + n^2) = (2n+1) \cdot \binom{n+1}{2}$$

No. you have to count the <u>same</u> collection two ways. Not two different collections.

□

8. Closing Statements

In all, mathematical proofs are quintessential elements that gives definite answers to if a proof is a true or untrue statement. The proofs above explained the various ways a proof can be written, and all have been proven to be true. Overall, the Introduction to Proofs has made me think drastically different when approaching a problem. Also, this class has and will help me in my future problems, either mathematically or rationally, and how to solve them.

Fin.

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